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Clamping pipelines

Abstract

A method of fixing a clamp around a pipe comprises supporting the pipe between curved clamp parts, namely a cradle and shells, that are assembled in mutual opposition about the pipe while fabricating the clamp. By placing the pipe into the cradle and then bringing the shells into contact with the pipe, those opposed clamp parts are simultaneously in contact with the pipe. Initially, a gap is maintained at an interface between the clamp parts. A full penetration weld is then formed in the gap. As the weld cools and shrinks, the clamp parts are drawn together, to a mutual spacing narrower than the original gap, to compress the pipe between them. The clamp is suitable for attaching the pipe to a structure such as the frame of a subsea pipeline accessory, for example a valve in fluid communication with the pipe.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4253779	12/1980	Grognu	405/188	F16L 1/26
4315702	12/1981	Moe	405/169	F16L 1/26
4459065	12/1983	Morton	166/347	F16L 1/26
5458439	12/1994	Hall	405/188	F16L 1/26
7942608	12/2010	Bastard et al.	N/A	N/A
8882392	12/2013	Cruz et al.	N/A	N/A
10371288	12/2018	Crisinelis	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108397597	12/2017	CN	F16L 1/20
113464723	12/2020	CN	N/A
2 250 415	12/2008	EP	N/A
2 510 270	12/2011	EP	N/A
1 305 516	12/1972	GB	N/A
2002 243068	12/2021	JP	N/A

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Background/Summary

(1) This invention relates to clamps for fixing pipelines to structures, for example anchor clamps for securing a subsea pipeline to an accessory structure. Conversely, the invention also relates to clamps for fixing items to pipelines, such as hold-back anchor fittings. In each case, there is a need to ensure that a pipeline cannot slip relative to a supporting structure or that a smaller item cannot slip relative to a supporting pipeline.

(2) In this specification, the invention will be described in the context of subsea pipelines or flowlines that lie on the seabed to convey hydrocarbons and other fluids from, or to, a subsea wellhead. References to pipelines herein are intended to incorporate risers that extend from the seabed to the surface to convey hydrocarbons and other fluids to, or from, a surface installation.
(3) Clamps are often fitted around pipelines in order to attach the pipeline to a structure or to attach an item to the pipeline. In each case, the clamp has to be fixed immovably to the pipeline.
(4) One example of a clamp is an anchor clamp that is mounted on a subsea structure connected to a subsea pipeline. Such a structure may, for example, be the foundation or frame of a pipeline accessory. The anchor clamp transfers mechanical loads between the pipeline and the structure. In particular, axial loads and bending moments in the pipeline are transferred to the structure during installation and operation.
(5) In this respect, subsea pipelines are routinely fitted with accessories during installation to provide operational flexibility, to create desired field layouts and to support future field extensions. Such accessories may be disposed at the ends of a pipeline and at in-line positions along its length. They include in-line tee assemblies (ILTs or ITAs), pipeline end manifolds (PLEMs), pipeline end terminations (PLETs, sometimes called flowline end terminations or FLETs), tie-in branches, valves, connectors, wyes, tees, shutdown valves, pigging connections, pig-launching equipment and pig-receiving equipment.
(6) Structures in line with the pipeline may therefore surround or support sensitive components such as tees or valves that have to be isolated from loads applied through the pipeline. For example, an anchor flange may be welded to or incorporated into the pipeline to transfer loads from the pipeline to a frame or foundation that supports such components. Alternatively, a bolted clamp around the pipeline may be fixed to the frame or foundation that supports such components.
(7) Another example of a clamp is a hold-back clamp, which may be attached to a subsea pipeline to restrain movement or ‘walking’ of the pipeline across the seabed driven by sea dynamics or by cycles of thermal elongation and contraction. Uncontrolled displacement of a subsea pipeline can cause the pipeline to buckle, can displace or damage structures connected to the pipeline and can make connections to other subsea infrastructure problematic. Thus, pipeline motion may be restrained by anchoring the pipeline as exemplified in U.S. Pat. No. 8,882,392, in which anchor chains connect hold-back clamps to anchors embedded in the seabed.
(8) A conventional pipeline clamp comprises two half-tubular portions or shells that are bolted together in mutual opposition around the pipeline to define a bore that receives the pipeline. The shells clamp together to engage an outer surface or coating of the pipeline so as to prevent relative movement between the clamp and the pipeline.
(9) In clamps that rely upon friction to hold the pipeline, resistance to movement is proportional to the area of contact between the inner surfaces of the shells and the outer surface of the pipeline. Thus, to ensure sufficient grip on the pipeline, friction clamps tend to be lengthy so as to maximise the area of contact. This increases their bulk and their cost. Also, the length and rigidity of a clamp may generate stress in the pipeline by restricting the freedom of the pipeline to bend along its length during installation and in use.
(10) Bolted clamps can be difficult to put in place and may present some risk for the pipeline, especially when located deep underwater. For example, due to issues such as creep, bolted connections between shells may not be able to maintain clamping force consistently throughout the long design life of a subsea pipeline. In principle, bolted friction clamps could be oversized to compensate for such deterioration but increasing the size of the clamp would worsen the related disadvantages noted above.
(11) Another conventional fastening method for a clamp employs a stopper or collar plate as described in EP 2250415. Alternatively, EP 2510270 teaches grouting a clamp onto a pipeline, typically by injecting epoxy grout. Both solutions add cost and complexity.
(12) It is also possible to incorporate a fixing structure into a pipeline, which involves interposing a

forged piece between, and welding the piece to, successive lengths of pipe. However, a forged piece is costly to produce. Also, producing, testing and coating the critical welds between the forged piece and the adjoining pipes takes a long time and requires a specific welders' qualification. These factors also add cost and complexity.

(13) Against this background, the invention resides in a method of fixing a clamp around a pipe. The method comprises: supporting the pipe between clamp parts that are disposed in mutual opposition about the pipe, the pipe being supported in a cradle of part-circular cross-section, the cradle being one of the clamp parts; engaging the pipe, when supported in the cradle, with at least one shell of part-circular cross-section, the or each shell being another of the clamp parts; maintaining a gap within at least one interface between the clamp parts; forming a weld in the or each gap, the or each weld being formed between a support of a shell and an outrigger of the cradle at a location offset laterally from a longitudinal axis of the pipe; and cooling and shrinking the or each weld to draw together, and to compress the pipe between, the clamp parts.

(14) The opposed clamp parts may be in contact with the pipe before and during formation of the or each weld, which may preferably be a full penetration weld. The clamp may conveniently be fabricated by assembling the clamp parts around the pipe and then forming the or each weld.

(15) The or each weld may, for example, be formed in a direction toward or away from the longitudinal axis of the pipe.

(16) Where there are two or more welds, those welds may advantageously be formed on opposed sides of a longitudinal plane that bisects the pipe, for example at locations that are laterally outboard of the longitudinal plane. Such welds may be in mutual alignment along a common axis, which axis may include a chord of a circular cross-section of the pipe.

(17) Various weld sequences are possible. For example, two or more welds may be formed simultaneously, sequentially, in a common direction or in mutually-opposed directions, or intermittently in alternation.

(18) The inventive concept also embraces a corresponding pipe-clamping arrangement that comprises: a pipe supported between clamp parts that are disposed in mutual opposition about the pipe, the pipe being supported in a cradle of part-circular cross-section, the cradle being one of the clamp parts; at least one shell of part-circular cross-section, the or each shell being another of the clamp parts; and at least one interface between the clamp parts. The pipe is held in compression between the clamp parts by virtue of shrinkage of at least one weld, for example a full penetration weld, that joins the clamp parts at the or each interface. The, or each, weld is formed at an interface between a support of the, or each, shell and an outrigger of the cradle at a location offset laterally from a longitudinal axis of the pipe.

(19) The pipe-clamping arrangement of the invention may further comprise a structure to which the pipe is attached by the clamp parts. Such a structure may suitably comprise, support or protect an accessory in fluid communication with the pipe.

(20) In summary, the invention pre-stresses shells around the pipeline by virtue of shrinkage due to welding. This exerts a strong mechanical clamping effort on the pipeline without, disadvantageously, welding directly to the pipeline or relying on bolted connections.

(21) GB 1305516 discloses fitting a two-part sleeve around the joint of a lined pipe and forming a circumferential weld between the annular parts of the sleeve to effect compressive sealing of the joint. In particular, the sleeve compresses the joint axially and so ensures fluid tightness as the weld between the parts of the sleeve cools and shrinks.

(22) In contrast to the simplicity of the invention, the technique described in GB 1305516 is fundamentally as challenging as welding a forged piece into the pipeline as acknowledged above. Also, unlike the invention, GB 1305516 uses weld shrinkage to draw together adjoining lengths of the pipeline in an axial or longitudinal direction during its fabrication. Such movement is of no use to clamp a prefabricated pipeline, which the invention achieves by using a radially-inward component of movement driven by weld shrinkage to force an external clamp element against the

exterior of the pipeline.
(23) Embodiments of the invention implement a method to grip a clamp around a tubular element on a support. The method comprises: manufacturing clamp shells designed to leave a shrinkage gap; providing a tubular element such as a horizontal pipe on the support; welding the clamp shells on the support so that the clamp shells are in contact with the tubular element and the weld has a gap that allows weld shrinkage; and cooling down the weld so that it shrinks and the clamp shells compress the tubular element.
(24) One or more of the clamp shells may comprise a partial cylinder or tube to embrace an arc of the pipe. The support may comprise a cradle to receive the pipe and at least one lateral support for welding the clamp.
(25) In summary, the invention provides a technique for fixing a clamp securely around a pipe. That technique involves supporting the pipe between oppositely-curved clamp parts such as a cradle and shells. The clamp parts are assembled in mutual opposition about the pipe while fabricating the clamp. By, for example, placing the pipe into the cradle and then bringing the shells into contact with the pipe, those opposed clamp parts are simultaneously in contact with the pipe.
(26) Initially, with all of the clamp parts kept in contact with the pipe, a gap is maintained at an interface between the clamp parts before a weld is formed in the gap. As the weld cools and shrinks, the clamp parts are drawn together, to a mutual spacing narrower than the original gap, to compress the pipe between them.
(27) The clamp is suitable for attaching the pipe to a structure such as the frame of a subsea pipeline accessory, for example a valve in fluid communication with the pipe.

Description

(1) In order that the invention may be more readily understood, reference will now be made, by way of example, to the accompanying drawings in which:
(2) FIG. **1** is a perspective view of a subsea pipeline incorporating an in-line accessory, the accessory being supported by a structure comprising a mudmat foundation to which the pipeline is attached by a clamp of the invention;
(3) FIG. **2** is an enlarged detail view corresponding to Detail II of FIG. **1**;
(4) FIG. **3** is an exploded cross-sectional view through the clamp and pipeline shown in FIGS. **1** and **2**;
(5) FIG. **4** corresponds to FIG. **3** but shows the clamp assembled around the pipeline;
(6) FIG. **5** is an enlarged view of the clamp while its parts are being welded together around the pipeline; and
(7) FIG. **6** corresponds to FIG. **5** but shows the clamp now fully welded and applying compressive force inwardly against the exterior of the pipeline by virtue of cooling and shrinkage of the welds.
(8) Referring firstly to FIG. **1** of the drawings, a subsea pipeline **10** incorporates an in-line accessory **12**, such as an in-line tee or valve, in fluid communication with the pipeline **10**. The accessory **12** is supported by a structure **14** that includes a frame **16** disposed around the accessory **12** and a foundation, exemplified here by a mudmat **18**, for providing an interface with the seabed soil. Once landed on a soft seabed of sand or silt, the weight of the accessory **12** and the frame **16** is spread and supported by the mudmat **18**, which keeps the accessory **12** and the pipeline **10** stable.
(9) As the pipeline **10** will undergo cycles of thermal elongation and contraction due to temperature fluctuations between operation and shut-down, the pipeline **10** and the accessory **12** are susceptible to move parallel to the seabed. To avoid buckling of the pipeline **10** under compressive stress as a result, the accessory **12** and the frame **16** are decoupled from the mudmat **18** in this example. Specifically, the accessory **12** and the frame **16** can move longitudinally relative to the mudmat **18** on parallel longitudinally-extending rails **20**. In this way, the accessory **12** can move horizontally

relative to the seabed, in a direction parallel to the principal axis of the pipeline **10**, while the mudmat **18** remains stationary on the seabed.

(10) The frame **16** supports an anchor clamp **22** that is arranged to clamp around the pipeline **10** at a position offset longitudinally along the pipeline **10** from the accessory **12**. The anchor clamp **22** thereby transfers loads from the pipeline **10** to the frame **16** along a load path that bypasses the accessory **12** and that extends from the frame **16** to the seabed via the mudmat **18**.

(11) Referring now also to FIGS. **2**, **3** and **4**, the frame **16** comprises a parallel pair of longitudinally-extending tubes **24** that extend orthogonally to, and are joined by, a transverse bridge structure **26**. The anchor clamp **22**, the tubes **24** and the bridge structure **26** are each symmetrical about a central longitudinal plane **28**. The bridge structure **26** supports the anchor clamp **22** on the central plane **28**.

(12) The anchor clamp **22** comprises an upwardly-concave, longitudinally-extending, part-tubular trough or cradle **30** of half-circular cross section that is bisected by the central plane **28**. The anchor clamp **22** further comprises a pair of downwardly-concave, longitudinally-extending, part-tubular shells **32** that are opposed to the cradle **30** and are mutually spaced one each side of the central plane **28**. Thus, the cradle **30** and the shells **32** are clamp parts that are angularly spaced from each other around the pipeline **10** and together clamp the pipeline **10** between them when assembled as shown in FIG. **4**.

(13) Specifically, the cradle **30** is disposed at the centre of a horizontal plate **34** of the bridge structure **22** and divides the plate **34** into two halves in mirror relation. The halves of the plate **34** are therefore outriggers with respect to the cradle **30**. The plate **34** extends in a plane parallel to, or containing, the central longitudinal axis **36** of the pipeline **10** when the anchor clamp **22** is assembled around the pipeline as shown in FIG. **4**.

(14) The cradle **30** may conveniently be formed by cutting along a pipe that is then welded to the halves of the plate **34** to join those halves together. The shells **32** may also conveniently be formed by cutting along a pipe, most efficiently the same pipe as is cut to make the cradle **30**. Thus, the cradle **30** and the shells **32** have a common radius of curvature, being slightly greater than or equal to the outer radius of the pipeline **10**.

(15) In this example, the cradle **30** and the shells **32** are each lined on their concave inner surfaces with a resilient, elastomeric liner **38** of, for example, neoprene rubber, as best appreciated in the enlarged views of FIGS. **5** and **6**. The liner **38** conforms to the outer contour of the pipeline **10** when compressed and so improves grip of the anchor clamp **22** on the pipeline **10**.

(16) The shells **32** are supported by, and cantilevered toward each other from, respective supports **40** that are welded to respective halves of the plate **34**. Thus, the plate **34** serves as a platform for the supports **40** to hold the shells **32** in engagement with the pipeline **10**.

(17) FIGS. **5** and **6** best show that each support **40** comprises an upright longitudinal batten **42** extending along a respective shell **32**. The shell **32** and the batten **42** are both held by a pair of parallel braces **44** that are joined to each other by a stiffening panel **46**. The braces **44** lie in longitudinally-spaced parallel planes that extend orthogonally relative to the central longitudinal axis **36** of the pipeline **10** and the plane of the plate **34**. Conversely, the panel **46** lies in a plane parallel to that axis **36** and at an acute angle to the plane of the plate **34**.

(18) The braces **44** of the supports **40** each have a flat underside **48** that is arranged to lie on top of the plate **34**, in parallel relation, so as to facilitate welding along their mutual interface. In this respect, specific reference is made to FIGS. **5** and **6**.

(19) FIG. **5** shows the pipeline **10** received in the cradle **30** and the shells **32** lying upon, and bearing against, the pipeline **10** in opposition to the cradle **30** about the central longitudinal axis **36**. At this stage, the braces **44** of the supports **40** are held spaced from the plate **34**, for example by being set up on a supporting jig, not shown, that may be fixed to the frame **16**. This leaves a gap **50** between the undersides **48** of the braces **44** and the plate **34**, which gap **50** facilitates the formation of full-penetration butt welds **52** between the braces **44** and the plate **34**. Typically, the welds **52** are

formed by manual arc welding although automatic welding or other welding techniques would be possible instead.
(20) The welds **52** are in mutual opposition about the central longitudinal plane **28**. In this example, the welds **52** are straight and are in mutual alignment along a common axis that forms a chord of the circular cross-section of the pipeline **10**.
(21) One of the welds **52** is shown schematically in FIG. **5** being formed by a torch or electrode **54** during a welding pass, hence filling the gap **50** in the process to fuse the supports **40** onto the plate **34**.
(22) In the example shown in FIG. **5**, the weld **52** is being formed as the torch or electrode **54** moves in a direction away from the central plane **28** but it would be possible instead to form the weld **52** in the opposite direction, as the torch or electrode **54** moves toward the central plane **28**. This example also shows one of the supports **40** being welded to the plate **34** before welding the other support **40** to the plate **34**.
(23) Other welding sequences are possible. For example, it would be possible instead to weld both of the supports **40** to the plate **34** simultaneously using two or more torches or electrodes **54** at once. In that case, welding could be effected by moving torches or electrodes **54** in the same direction for both supports **40** or in mutually-opposed directions with respect to the central plane **28**, or intermittently in alternation between the supports **40**. Also, each weld **52** may be formed by performing passes of one or more torches or electrodes **54** on one or both sides of each brace **44**, simultaneously or in succession.
(24) FIG. **6** shows the welds **52** of both supports **40** now completed and cooled. As enabled by the gaps **50**, the welds **52** have shrunk and narrowed while cooling, thus pulling the supports **40** toward the plate **34** and hence pressing the shells **32** onto the pipeline **10** as shown. The pressure applied by the shells **32** pushes the pipeline **10** into the opposed cradle **30**, which exerts a reaction force against the pipeline **10** accordingly. The compression applied to the pipeline **10** by the cradle **30** and the shells **32** induces a reaction of hoop stress in the wall of the pipeline **10**.
(25) It will be apparent that the resilient liners **38** within the cradle **30** and the shells **32** are compressed and hence thinner in FIG. **6** relative to their greater thickness in the relatively relaxed state shown in FIG. **5**.
(26) By virtue of shrinkage of the welds **52**, the increased inward pressure applied by the cradle **30** and the shells **32** against the pipeline **10** increases the frictional grip of the anchor clamp **22** on the pipeline **10**. Elegantly, this effect is achieved by virtue of the fabrication process and without requiring the anchor clamp **22** to be enlarged, or for any parts to be bolted together, or for any welding to be performed on the pipeline **10** itself. The result is a simple, quick, consistent and effective solution to the problem of fixing pipelines to structures or, conversely, of fixing items to pipelines.
(27) Natural cooling of the welds **52** may be preferred to forced cooling, for example by air blowing, water spraying or quenching. This is because forced cooling could introduce defects or unbalance the contact pressure on the pipeline **10** if cooling is not consistent between the welds **52** or along the length of the welds **52**.
(28) Many other variations are possible within the inventive concept. For example, some mudmat foundations are designed to slide with an accessory relative to the seabed, and so need not provide for the accessory to move relative to the mudmat. Foundations other than mudmats, such as piles embedded in the seabed, are also possible. An anchor clamp of the invention may be provided on structures other than foundations, and on smaller items that are to be fixed to pipelines.
(29) There could be one shell, for example extending laterally to both sides of the central longitudinal plane, or more than two shells.

Claims

1. A method of fixing a clamp around a pipe, the method comprising: supporting the pipe between clamp parts that are disposed in mutual opposition about the pipe, in a cradle of part-circular cross-section, the cradle being one of the clamp parts; engaging the pipe, when supported in the cradle, with at least one shell of part-circular cross-section, the or each shell being another of the clamp parts; maintaining a gap within at least one interface between the clamp parts; forming a weld in the or each gap, the or each weld being formed between a support of a shell and an outrigger of the cradle at a location offset laterally from a longitudinal axis of the pipe; and cooling and shrinking the or each weld to draw together, and to compress the pipe between, the clamp parts.
2. The method of claim 1, comprising contacting the pipe with the opposed clamp parts before and during formation of the or each weld.
3. The method of claim 1, wherein the or each weld is a full penetration weld.
4. The method of claim 1, comprising fabricating the clamp by assembling the clamp parts around the pipe and then forming the or each weld.
5. The method of claim 1, comprising forming the or each weld in a direction toward or away from the longitudinal axis of the pipe.
6. The method of claim 1, comprising forming welds on opposed sides of a longitudinal plane that bisects the pipe.
7. The method of claim 6, comprising forming the welds at locations that are laterally outboard of the longitudinal plane.
8. The method of claim 6, wherein the welds are in mutual alignment along a common axis.
9. The method of claim 8, wherein the common axis includes a chord of a circular cross-section of the pipe.
10. The method of claim 6, comprising forming the welds simultaneously.
11. The method of claim 6, comprising forming the welds intermittently in alternation.
12. The method of claim 6, comprising forming the welds in a common direction.
13. The method of claim 6, comprising forming the welds in mutually-opposed directions.
14. A pipe-clamping arrangement, comprising: a pipe supported between clamp parts that are disposed in mutual opposition about the pipe, wherein the pipe is supported in a cradle of part-circular cross-section, the cradle being one of the clamp parts; at least one shell of part-circular cross-section, the or each shell being another of the clamp parts; and at least one interface between the clamp parts; wherein the pipe is in compression between the clamp parts by virtue of shrinkage of at least one weld that joins the clamp parts at the or each interface, and wherein the or each weld is formed at an interface between a support of a shell and an outrigger of the cradle at a location offset laterally from a longitudinal axis of the pipe.
15. The arrangement of claim 14, wherein the or each weld is a full penetration weld.
16. The arrangement of claim 14, comprising welds on opposed sides of a longitudinal plane that bisects the pipe.
17. The arrangement of claim 16, wherein the welds are at locations that are laterally outboard of the longitudinal plane.
18. The arrangement of claim 16, wherein the welds are in mutual alignment along a common axis.
19. The arrangement of claim 18, wherein the common axis includes a chord of a circular cross-section of the pipe.
20. The arrangement of claim 14, further comprising a structure to which the pipe is attached by the clamp parts.
21. The arrangement of claim 20, wherein the structure comprises an accessory in fluid communication with the pipe.